Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Select the correct IUPAC name for $[CoCl_2(en)_2]_2\ (ClO_3)_2$.

(A) Dichloridobis (ethylenediamine)cobalt(III) chlorate

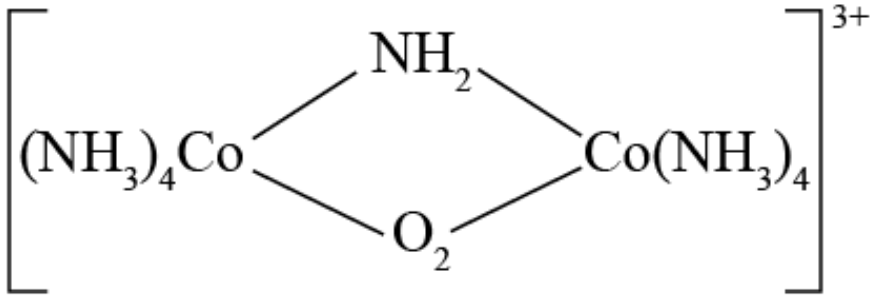
(B) Dichloridobis (ethane-1, 2 -diamine)cobalt(III) chlorate

(C) bis {dichloridoethylenediaminecobalt(III)} chlorate

(D) bis {di(chlorido)ethylenediaminecobalt(III)} chlorate

Q.2

select the correct IUPAC name for the following:



(A) Tetramminecobalt (III)- μ-amido- μ-peroxido tetraamminedicobalt (III) ion

(B) μ-Amido- μ-peroxidobis(tetraammine)dicobalt-(III) ion

(C) μ-Amido- μ-peroxidobis(tetraamminecobalt(III)) ion

(D) μ-Amido- μ-peroxidooctaamminedicobalt(III) ion

Q.3

Select the correct IUPAC name for $[Mo\ (\sigma - C_3H_5)\ (\pi - C_5H_5)\ (CO)_3]$

(A) Tricarbonyl (η^5-cyclopentadienyl) allyl molybdate(II)

(B) Allytricarbonyl $\left(\eta^5\right.$ -cyclopentadiene) molybdate(II)

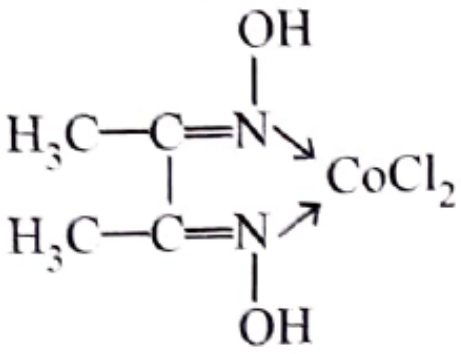
(C) Allyltricarbonyl $\left(\eta^5\right.$ -cyclopentadiene) molybdenum(II)

(D) Allyltricarbonyl $\left(\eta^5 -\right.$ cyclopentadienyl) molybdenum(II)

Q.4

@IITJEE_Advanced

The correct IUPAC name of the complex:



is

(A) Dichloridodimethylglyoximatocobalt(II)

(B) Bis(dimethylglyoxime) dichlorocobalt(II)

(C) Dimethylglyoximecobalt(II) chloride

(D) Dichloridodimethylglyoxime-N,N-cobalt(II)

Q.5

Coordination number and oxidation state of Cr in $\mathrm{K_3\left[Cr\left(C_2O_4\right)_3\right]}$ are, respectively:

(A) 3 and $+3$

(B) 3 and 0

(C) 6 and $+3$

(D) 4 and $+2$

Q.6

What is not true about $O^{-}\mathrm{COCOO}^{-}$?

(A) It is called oxalate

(B) It can produce chelation

(C) It is symmetrical bidentate ligand

(D) It is a tetradentate ligand

Q.7

Which of the following is not a complex salt ?

(A) $\mathrm{K_4\left[Fe(CN)_6\right]}$

(B) $\mathrm{\left[Co\left(NH_3\right)_6\right]Cl_3}$

(C) $\mathrm{\left(NH_4\right)_2SO_4, Fe_2\left(SO_4\right)_s .24H_2O}$

(D) $\mathrm{K_2\left[Cd(CN)_4\right]}$

@IITJEE_Advanced

Q.8

Which is not π - bonded complex ?

(A) Zeise's salt

(B) Ferrocene

(C) Dibenzene chromiun

(D) Tetraethyl lead

Q.9

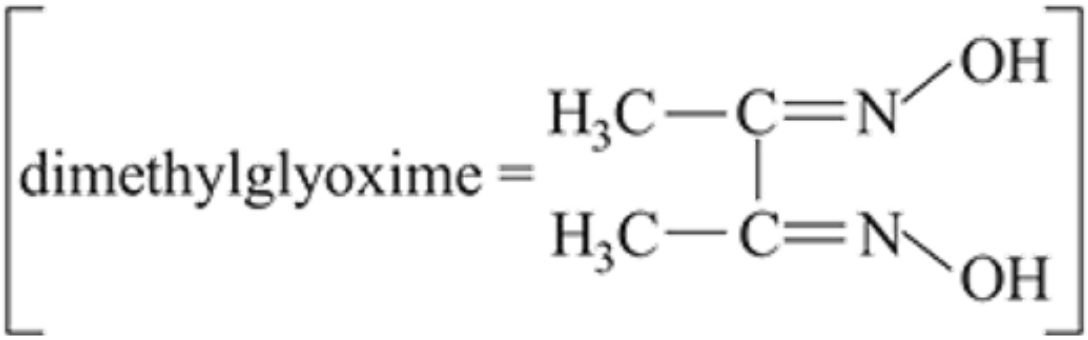
Red precipitate is obtained when ethanol solution of dimethylglyoxime is added to ammoniacal Ni(II). Which of the following statements is not true?

(A) Red complex has a square planar geometry.

(B) Complex has symmetrical H-bonding

(C) Red complex has a tetrahedral geometry.

(D) Dimethylglyoxime functions as bidentate ligand.



Q.10

The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1, L_2, L_3 and L_4 absorb wavelengths in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is:

(A) $L_4 < L_3 < L_2 < L_1$

(B) $L_1 < L_3 < L_2 < L_4$

(C) $L_3 < L_2 < L_4 < L_1$

(D) $L_1 < L_2 < L_4 < L_3$

Q.11

Which one of the following coordination compounds is used to inhibit the growth of tumours?

@IITJEE_Advanced

(A) Trans-platin

(B) EDTAcomplex of calcium

(C) $[(Ph_3P)_3\,RhCl]$

(D) Cis-platin

Q.12

The EAN of iron in potassium ferricyanide is

(A) 18

(B) 54

(C) 35

(D) 23

Q.13

AgCl is soluble in NH_4OH solution. The solubility is due to formation of

(A) $AgOH$

(B) Ag_2O

(C) $[Ag\,(NH_3)_2]^+$

(D) NH_4Cl

Q.14

In which of the following coordinate compounds the central metal atom obeys the EAN rule.

(A) $K_3\,[Fe(CN)_6]$

(B) $K_4\,[Fe(CN)_6]$

(C) $[Cu\,(NH_3)_4]\,SO_4$

(D) All of these

Q.15

Which of the following will exhibit maximum ionic conductivity?

(A) $K_4\,[Fe(CN)_6]$

(B) $[Co\,(NH_3)_6]\,Cl_3$

(C) $[Cu\,(NH_3)_4\,Cl_2]$

(D) $[Ni(CO)_4]$

@IITJEE_Advanced

Q.16

Which of the following complexes have a maximum number of unpaired electrons?

(A) $[Ni(CO)_4]$

(B) $[Co\,(NH_3)_4\,(NO_2)_2]^+$

(C) $[Ag(CN)_2]^-$

(D) $[CuBr_4]^{2-}$

Q.17

The degeneracy of d -orbitals is lost under
(I) Strong field ligand
(II) Weak field ligand
(III) Mixed field ligand
(IV) Chelated ligand field

(A) I, II and IV

(B) I and II

(C) I, II, III and IV

(D) I, II and III

Q.18

IUPAC name of $[Pt\,(NH_3)_3\,(Br)\,(NO_2)\,Cl]\,Cl$ is

(A) Triamminechlorobromonitroplatinum (IV) chloride

(B) Triammincbromonitrochloroplatimum (IV) chloride

(C) Triamminebromochloronitroplatinum (IV) chloride

(D) Triamminenitrochlorobromoplatinum (IV) chloride

Q.19

@IITJEE_Advanced

Which of the following name formula combinations is not correct?

	Formula	IUPAC Name
(a)	$K_2[Pt(CN)_4]$	Potassium tetracyanoplatinate (II)
(b)	$[Mn(CN)_5]^{2-}$	Pentacyanomagnate (II) ion
(c)	$K[Cr(NH_3)_2Cl_4]$	Potassium diammine tetrachlorochromate (III)
(d)	$[Co(NH_3)_4(H_2O)I]SO_4$	Tetraammine aquaiodo cobalt(III) sulphate

(A) option a

(B) option b

(C) option c

(D) option d

Q.20

In $[Co\ (NH_3)_6]\ Cl_3$, the number of covalent bonds is

(A) 3

(B) 6

(C) 9

(D) 18

Answers & Solutions

Q.1 Answer:

Dichloridobis (ethylenediamine)cobalt(III) chlorate

Solution:

Dichloridobis (ethylenediamine)cobalt(III) chlorate

Q.2 Answer:

μ-Amido- μ-peroxidobis(tetraammine)dicobalt-(III) ion

Solution:

μ-Amido- μ-peroxidobis(tetraammine)dicobalt-(III) ion

Q.3 Answer:

@IITJEE_Advanced

Allyltricarbonyl $\left(\eta^5-\right.$ cyclopentadienyl) molybdenum(II)

Solution:

The correct I.U.P.A.C name for $\mathrm{Mo}\left(\sigma-\mathrm{C_3H_5}\right)\left(\pi-\mathrm{C_5H_5}\right)\left(\mathrm{CO}\right)_3$:
is Allytricarbonyl (η^5 - cyclopentadienyl) molybdenum (II).
The central atom is Mo in + oxidation state which is represented by the roman numerals II in the pranthesis.
3 carbonyl ligands, one allyl ligand and one cyclopentadienyl ligand is present.
η^5 represents that the cyclopnetadienyl ligand shares 5 electrons with metal.
Hence,option D is correct.

Q.4 Answer:

Dichloridodimethylglyoximatocobalt(II)

Solution:

Dichloridodimethylglyoximatocobalt(II)

Q.5 Answer:

6 and $+3$

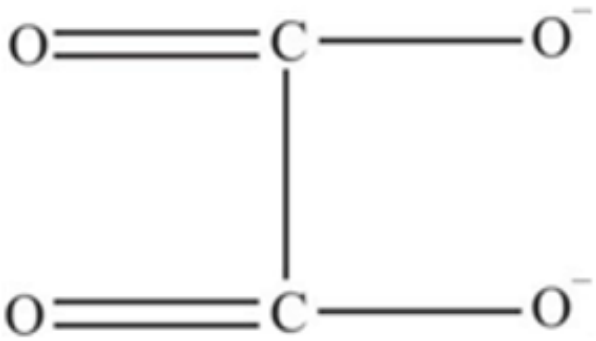
Solution:

As oxalate is a bidentate ligand, coordination number of Cr is 6 .
Since the overall charge on the complex is 0 , the sum of oxidation states of all elements in it should be equal to 0
Let the oxidation number of Cr be x.
Therefore, $3+x+3\times(-2)=0$
or, $x=+3$

Q.6 Answer:

It is a tetradentate ligand

Solution:



$\rightarrow$ oxalate ion is bidentate due to two donor atom $\left(\mathrm{O}^-\right)$

@IITJEE_Advanced

Q.7 Answer:

$(NH_4)_2\,SO_4, Fe_2\,(SO_4)_s\,.24H_2O$

Solution:

$(NH_4)_2\,SO_4 \cdot Fe_2\,(SO_4)_3 \cdot 24H_2O \rightarrow (NH_4)_2\,SO_4 + Fe_2\,(SO_4)_3 + 24H_2O$
$2NH_4^+ + SO_4^- + 2Fe^{3+} + 3SO_4 + 24H_2O$

Q.8 Answer:

Tetraethyl lead

Solution:

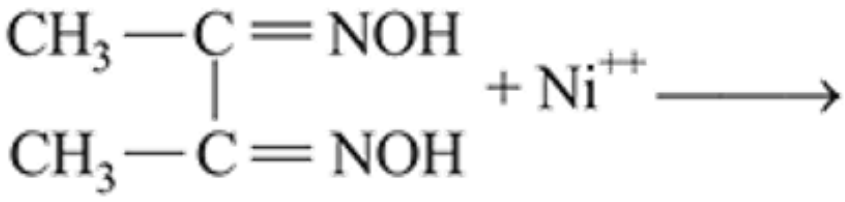
Tetraethyl lead $Pb\,(C_2H_5)_4$ is not π bonded complex. It is σ bonded organometallic compound

Q.9 Answer:

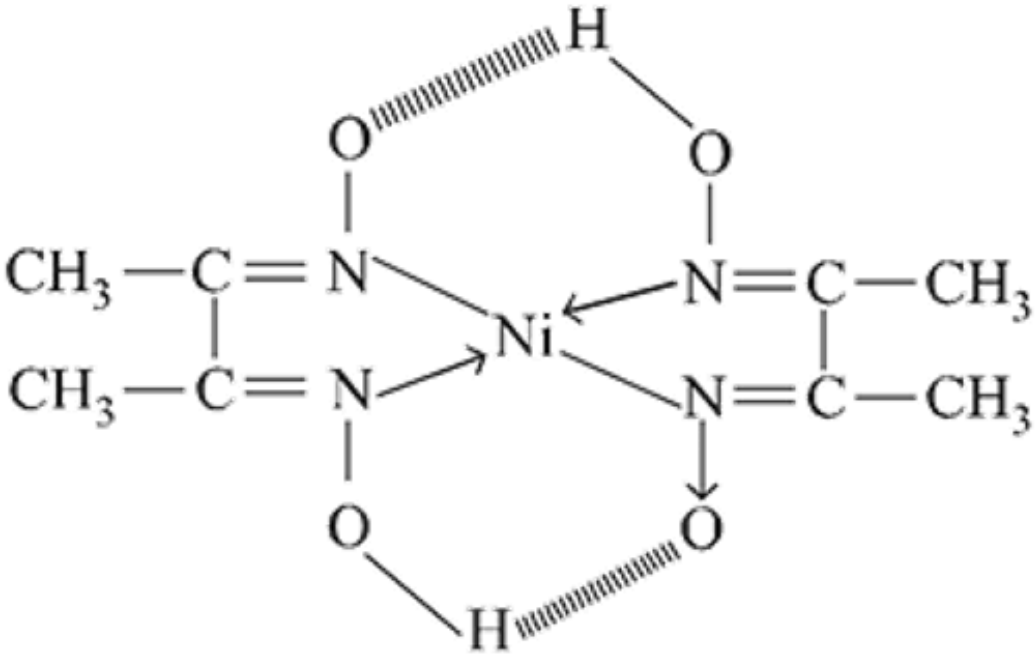
Red complex has a tetrahedral geometry.

Solution:

Nickel ions are frequently detected by the formation of red precipitate of the complex of nickel dimethylglyoxime, when heated with dimethylglyoxime.



Dimethylglyoxime



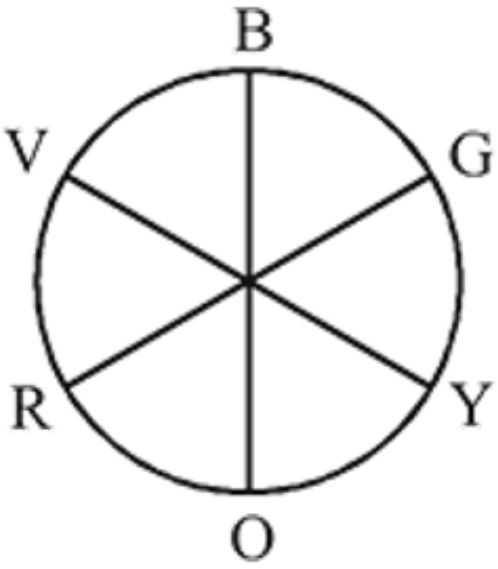
Nickel dimethylglyoxime

Q.10 Answer:

@IITJEE_Advanced

$L_1 < L_3 < L_2 < L_4$

Solution:



For a given metal ion, weak field ligands create a complex with smaller Δ, which will absorbs light of longer λ and thus lower frequency. Conversely, stronger field ligands create a larger Δ, absorb light of shorter λ and thus higher v i.e. higher energy.

$$\underset{\lambda=650\text{ nm}}{\text{Red}} < \text{Yellow 570 nm} < \underset{490\text{ nm}}{\text{Green}} < \text{Blue}$$

So order of ligand strength is

$$L_1 < L_3 < L_2 < L_4$$

Q.11 Answer:

Cis-platin

Solution:

The design of metal-based anticancer drugs was initiated by cisplatin (cis-Pt(NH3)2Cl2), one of the leading agents in clinical use with a high cytotoxic effect upon a variety of tumour types, in particular against testicular, ovarian, head and neck, and bladder carcinomas, as well as lymphoma.

Q.12 Answer:

35

Solution:

EAN of a central metal ion $=$ (atomic no. of central atom) $-$ oxidation state $+$ no. of ligands

$\times 2 = 26 - 3 + (6 \times 2) = 23 + 12 = 35$

@IITJEE_Advanced

Q.13 Answer:

$[Ag(NH_3)_2]^+$

Solution:

AgCl form soluble complex with NH_4OH
which is $[Ag(NH_3)_2]^+$

Q.14 Answer:

$K_4[Fe(CN)_6]$

Solution:

$K_4[Fe(CN)_6]$ **obeys the EAN rule.**

Q.15 Answer:

$K_4[Fe(CN)_6]$

Solution:

$K_4[Fe(CN)_6]$ **in solution furnishes** $4K^+ +$
$[Fe(CN)_6]^4$ **five ions and exhibits maximum ionic conductivity.**

Q.16 Answer:

$[CuBr_4]^{2-}$

Solution:

$Ni(CO)_4; \mu_{eff} = 0$
$[Co(NH_3)_4(NO_2)_2]^+; \mu_{eff} = 0$
$[Ag(CN)_2]; \mu_{eff} = 0$
$[CuBr_4]^{2-}; \mu_{eff} = 1.732BM$

Q.17 Answer:

I, II, III and IV

Solution:

Degenerate d **-orbitals undergo splitting under ligand field created by strong, weak or mixed ligands.**

Q.18 Answer:

Triamminebromochloronitroplatinum (IV) chloride

Solution:

IUPAC name $[Pt(NH_3)_3(Br)(NO_2)Cl]Cl$
$x + 1(-1) + 1(-1) = +1$
$x = +4(IV)$

@IITJEE_Advanced

$Br^- \rightarrow$ **Bromo**

$NO_2^- \rightarrow$ **Nitro**

$Cl^- \rightarrow$ **chloro**

$NH_3 \rightarrow$ **Ammine**

Triammine bromo chloronitroplatinum (IV) chloride these ligands name are in alphabetical order (either the ligand is ionic or neutral)

Q.19 Answer:

option b

Solution:

The correct IUPAC Name of $[Mn(CN)_5]^{2-}$ Pentacyanomangnate (III).

Q.20 Answer:

18

Solution:

The number of ammonia molecules is 6 . Each ammonia molecule contains 3 covalent bonds between N and H. Therefore the number of covalent bonds is 18 .

@IITJEE_Advanced